

به نام خدا

تحلیل کار و اندازه پروژه

تهیه توسط سجاد (فرمول سلول ها با کمک AI)

توضیحات:

نکته: فقط اعداد آبی به عنوان ورودی این جداول در نظر گرفته شده اند. سایر اعداد یا ثابت اند، یا حاصل عملیات ریاضی هستند.

اکتورهای انسانی:

- Org Admin

- Org Staff

- Super Admin

- Org Support

- Customer

سیستم های خارجی که با پروژه ارتباط برقرار میکنند.

- OTP (ارتباط از طریق API)

- CloudFlare (ارتباط از طریق HTTP)

- سیستم برای هاست دیتابیس MSSQL (ارتباط TCP/IP)

نکته: هاست بک اند را به حساب نمی آوریم زیرا مرجع سامانه است که با سیستم های خارجی ارتباط می گیرد.

از این فرمول برای PHM استفاده شده.

```
If the sum of (number of Efactors E1 through E6 assigned value < 3) and
(number of Efactors E7 and E8 assigned value> 3)
    ≤ 2
    PHM = 20
Else If the sum of (number of Efactors E1 through E6 assigned value < 3)
and
    (number of Efactors E7 and E8 assigned value> 3)
    = 3 or 4
    PHM = 28
Else
    Rethink project; it has too high of a risk for failure
```

در نتیجه:

$$\Rightarrow PHM = 20$$

Unadjusted Actor Weighting Table:

Actor Type	Description	Weighting Factor	Number	Result
Simple	External system with well-defined API	1	1	1
Average	External system using a protocol-based interface: HTTP, TCP/IP, database	2	2	4
Complex	Human	3	5	15
Unadjusted Actor Weight Total (UAW)				20

Unadjusted Use-Case Weighting Table

Use-Case Type	Description	Weighting Factor	Number	Result
Simple	1-3 transactions	5	13	65
Average	4-7 transactons	10	4	40
Complex	> 7 transactions	15	3	45
Unadjusted Use-Case Weighting Total (UUCW)				150
Unadjusted Use-Case Points (UUCP) = UAW + UUCW		=	170	

Technical Complexity Factors:

Factor Number	Descriptions	Weight	Assigned Value(0-5)	Weighted Value	Notes
T1	Distributed system	2.0	3	6	
T2	Response time or throughput performance objectives	1.0	2	2	
T3	End-user online efficiency	1.0	4	4	
T4	Complex internal processing	1.0	2	2	
T5	Reusability of code	1.0	3	3	
T6	Ease of installation	0.5	2	1	
T7	Ease of use	0.5	4	2	
T8	Portability	2.0	5	10	
T9	Ease of change	1.0	4	4	
T10	Concurrency	1.0	4	4	
T11	Special security objectives included	1.0	3	3	
T12	Direct access for third parties	1.0	0	0	
T13	Special user training required	1.0	0	0	
Technical Factor Value (TFactor)				41	
Technical Complexity Factor:(TCF) = $0.6 + (0.01 * TFactor)$				1.01	

Environmental Factors:

Factor Number	Descriptions	Weight	Assigned Value(0-5)	Weighted Value	Notes
E1	Familiarity with system development process being used	1.5	3	4.5	
E2	Application experience	0.5	2	1	
E3	Object-oriented experience	1.0	4	4	
E4	Lead analyst capability	0.5	4	2	
E5	Motivation	1.0	5	5	
E6	Requirements stability	2.0	3	6	
E7	Part-time staff	-1.0	5	-5	
E8	Difficulty of programming language	-1.0	3	-3	
Environmental Factor Value (EFactor)				14.5	
Environmental Factor (EF) = $1.4 + (-0.03 * EFactor)$				0.965	
Adjusted Use-Case Points (UCP) = $UUCP * TCF * EF$				165.6905	
Person-Hour Multiplier(PHM)				20	
Effort in Person Hours = $UCP * PHM$				3313.81	